

Internship Project — Week 1

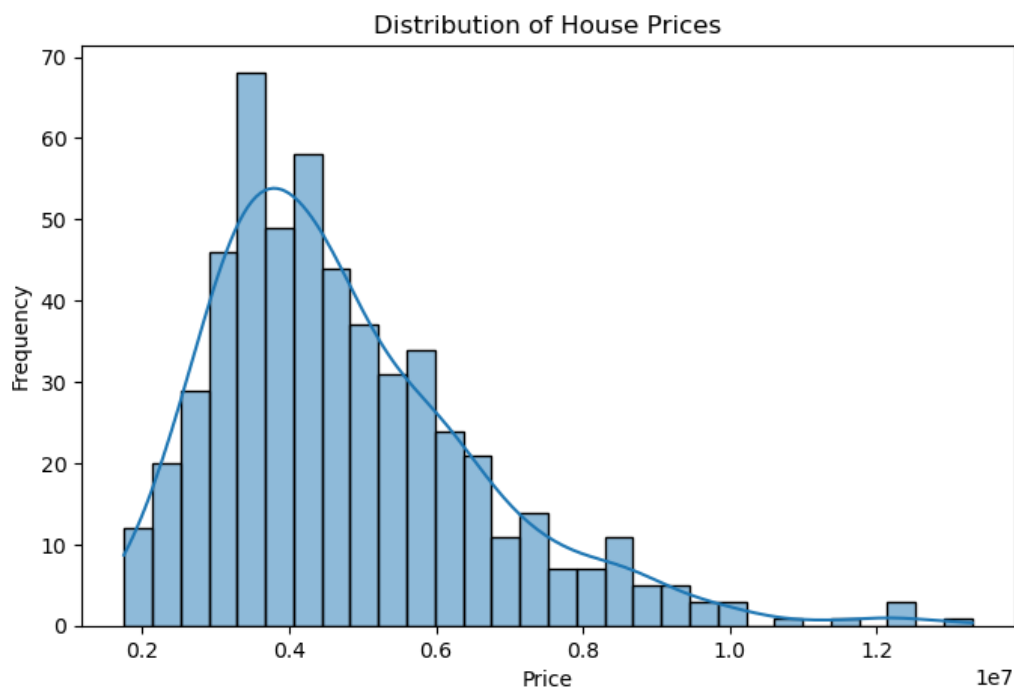
Project Title: House Price Prediction

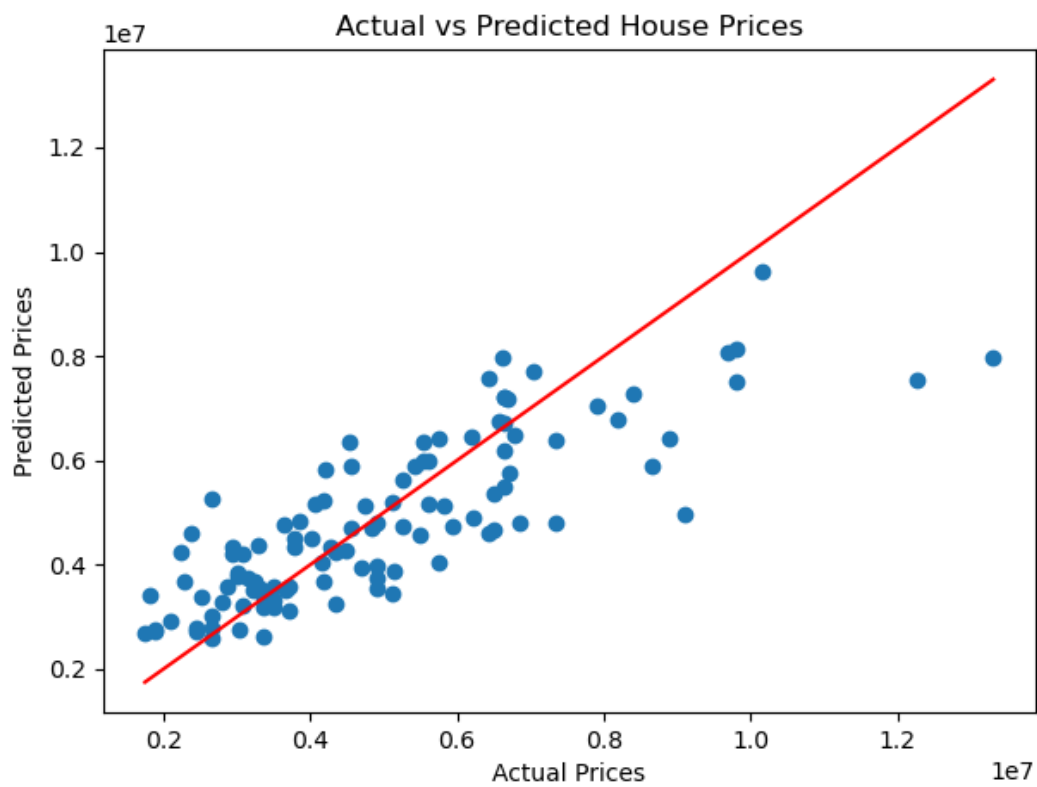
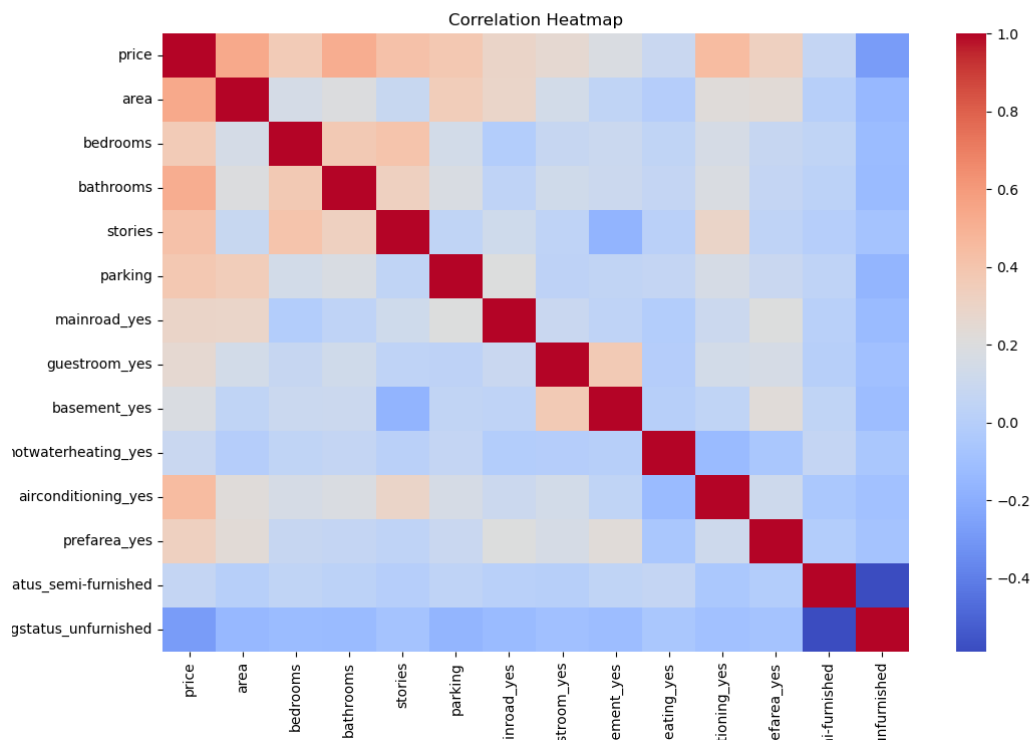
Submitted by: Indrayani Prashant Mude

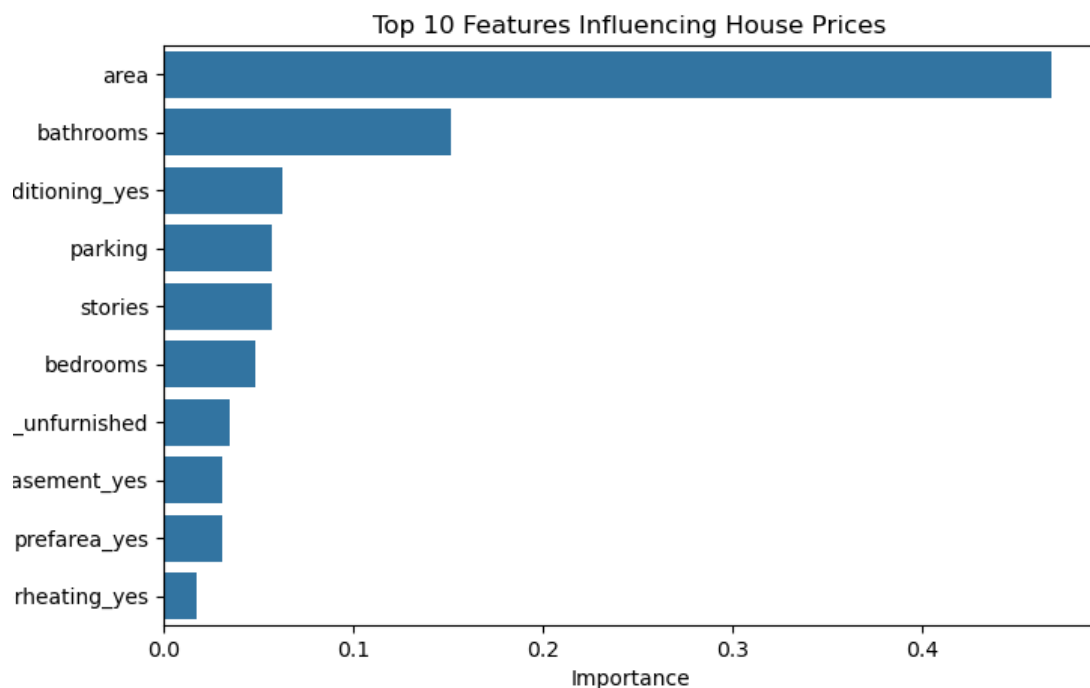
St. Vincent Pallotti College of Engineering and Technology, Nagpur

Problem Statement: Real estate buyers and sellers often rely on guesswork or outdated comparisons to estimate a property's fair value. Your task is to build a regression model that predicts house prices based on property features such as size, number of rooms, location, and age — and then identify which features most strongly influence price.

Graphs:







Summary

The goal of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics.

After analyzing the dataset and training two regression models, it was found that **Linear Regression outperformed Random Forest Regressor**, achieving an R^2 score of approximately **0.65**, which indicates that the model could explain about 65% of the variation in house prices.

Feature importance analysis showed that **area** was the most significant factor influencing house prices, followed by **number of bathrooms, air conditioning, parking availability, and number of stories**. One interesting observation was that the size of the property had a much stronger impact on price than several other amenities.

The Actual vs Predicted scatter plot demonstrated that the model was able to capture the overall pricing trend reasonably well, although prediction errors increased for some high-priced properties.

Based on these findings, real estate businesses should pay particular attention to property size and essential amenities when estimating prices, valuing properties, or designing marketing strategies.